

COMPUTER VISION

Computer vision is concerned with the theory for building artificial systems that obtain information from images. The image data can take many forms, such as a video sequence, views from multiple cameras, or multi-dimensional data from a medical scanner.

Typical Applications of computer vision

1. *Recognition*: The classical problem in computer vision, image processing and machine vision is that of determining whether or not the image data contains some specific object, feature, or activity.
2. *Motion*: Several tasks relate to motion estimation, in which an image sequence is processed to produce an estimate of the velocity or to track objects
3. *Scene reconstruction*: Given one or (typically) more images of a scene, or a video, scene reconstruction aims at computing a 3D model of the scene.
4. *Image restoration*: The aim of image restoration is the removal of noise (sensor noise, motion blur, etc.) from images using various filters

Functions Computer vision systems

Typical functions which are found in many computer vision systems.

1. **Image acquisition**: A digital image is produced by one or several image sensor which, besides various types of light-sensitive cameras, includes range sensors, tomography devices, radar, ultra-sonic cameras, etc.
2. **Pre-processing**: Before a computer vision method can be applied to image data in order to extract some specific piece of information, it is usually necessary to process the data in order to assure that it satisfies certain assumptions implied by the method. Examples are Noise reduction and Contrast enhancement
3. **Feature extraction**: Image features such as lines and edges are extracted from the image data.
4. **Detection/Segmentation**: A decision is made about which image points or regions of the image are relevant for further processing. For example segmentation of one or multiple image regions which contain a specific object of interest.
5. **High-level processing**: It deals with classifying a detected object into different categories and estimation of application specific parameters, such as object pose or object size.

MACHINE VISION

Machine vision (MV) is the application of computer vision to industry and manufacturing. Whereas computer vision is mainly focused on machine-based image processing, machine vision most often requires also digital input/output devices and computer networks to control other manufacturing equipment such as robotic arms. One of the most common applications of Machine Vision is the inspection of manufactured goods such as semiconductor chips, automobiles, food and pharmaceuticals. machine vision systems use digital cameras, smart cameras and image processing software to perform visual inspection of parts in factory

assembly lines. Machine vision systems are programmed to perform narrowly defined tasks such as counting objects on a conveyor, reading serial numbers, and searching for surface defects. These systems must examine individual pixels of images, and process them to develop conclusions with the assistance of knowledge bases and features such as pattern recognition engines.

Components of a machine vision system

A typical machine vision system will consist of several among the following components:

1. One or more digital or analog camera (black-and-white or colour) with suitable optics for acquiring images
2. Camera interface for digitizing images (widely known as a "frame grabber")
3. A processor (often a PC or embedded processor, such as a DSP)
4. Input/Output hardware (e.g. digital I/O) or communication links (e.g. network connection or RS-232) to report results
5. Lenses to focus the desired field of view onto the image sensor.
6. Suitable, often very specialized, light sources (LED illuminators, fluorescent or halogen lamps etc.)
7. A program to process images and detects relevant features.
8. A synchronizing sensor for part detection (often an optical or magnetic sensor) to trigger image acquisition and processing.
9. Some form of actuators used to sort or reject defective parts.

The sync sensor determines when a part (often moving on a conveyor) is in position to be inspected. The sensor triggers the camera to take a picture of the part as it passes beneath the camera and often synchronizes a lighting pulse to freeze a sharp image. The lighting used to illuminate the part is designed to highlight features of interest and obscure or minimize the appearance of features that are not of interest (such as shadows or reflections).

The camera's image is captured by the frame grabber. A frame grabber is a digitizing device that converts the output of the camera to digital format and places the image in computer memory so that it may be processed by the machine vision software.

The software will typically take several steps to process an image. Often the image is first manipulated to reduce noise. Following the initial simplification, the software will count, measure, and/or identify objects, dimensions, defects or other features in the image. As a final step, the software passes or fails the part according to programmed criteria. If a part fails, the software may signal a mechanical device to reject the part; alternately, the system may stop the production line and warn a human worker to fix the problem that caused the failure.